Girls Build Bots — Junior STEM Career Exploration

Badge-aligned workshop guide for robotics-team-led outreach

Level: Juniors (Grades 4–5) · **Badge:** STEM Career Exploration **Format:** One 90–120 minute session · **Ratio:** 1 mentor per 4–5 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. This badge is the one where **the team itself is the main resource** — every mentor is a live example of a STEM path in progress. A robot demo is a wonderful bonus but not required.

> **Troop Leaders:** This curriculum is designed to *align with* the Junior STEM Career Exploration badge. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The big idea

Juniors are old enough to see that STEM careers are *many* — not just "scientist" — and that their own interests and skills already point somewhere. The session connects interests → skills → real careers, with the robotics team as living proof that the path starts young.

Vocabulary: career, skill, interest, engineer, computer science, role, pathway.

STEM Career Exploration (90–120 minutes)

Materials

- "Career stations" set up by team role (see Activity 3): a small prop/demo per station
- Interest-to-career matching sheet: What I like / Skills it uses / Careers that use those skills
- Team members prepped with 2-minute "my STEM story" talks
- Optional: robot for demos at the stations

Activity 1 — STEM Is Bigger Than You Think (15 min)

Rapid-fire: call out jobs, Scouts vote STEM or not — then reveal the STEM inside surprising ones (video-game music composer → audio technology; sneaker designer → materials engineering; zookeeper → animal science and data). **Point:** almost every field has a STEM path into it, so your interests already count.

Activity 2 — Team Panel: My STEM Story (25 min)

4–5 team members each give a 2-minute story: what they do on the team, how they started, one struggle, one thing they love. Prioritize a mix of roles — programmer, builder, designer,

strategist — and make sure Scouts hear from young women on the team. Then open Q&A. Near-peer role models are the most powerful element of this badge.

Activity 3 — Career Stations (35 min)

Scouts rotate through 4 stations run by team members, each mapping a team role to real careers, with a 5-minute hands-on taste: - **Programming station** → software engineer, roboticist (try a 5-line block program or trigger a robot action) - **Build/CAD station** → mechanical engineer, product designer (sketch a part, handle real components) - **Strategy/data station** → data analyst, engineer-manager (read a simple match-data chart, pick the best alliance) - **Media/outreach station** → technical writer, science communicator (write a one-line caption for a robot photo)

At each station Scouts jot on their matching sheet: did I like this? what skill did it use?

Activity 4 — My Pathway Map (15 min) + Wrap (10 min)

Each Scout completes her sheet: top interest → skills it uses → one career that fits → **one concrete next step** (join FLL, try a free coding course, interview a family friend in STEM, read one book). Closing circle: every Scout states her career pick and her step out loud — saying it makes it real.

> **Robot moment (optional):** End with a robot demo and let Scouts drive or trigger it — "I could do this" lands hardest with hands on the controls.

Completion Checklist — STEM Career Exploration

By the end of the session, to earn the badge, each Scout can: - [ ] Explain that STEM careers exist inside almost every field - [ ] Describe two different STEM careers after meeting people on those paths - [ ] Connect her own interests and skills to a career that uses them - [ ] Name one STEM career to explore further and one concrete next step

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Mentor Guide (for the robotics team)

- **You are the curriculum.** Prep your 2-minute story: how you started, a struggle, why you stay. Real beats polished.
- **Near-peer is the magic.** Say "I was in fourth grade once too, and nobody handed me a robot." It lands.
- **Center the young women on your team** at the panel and the stations — representation is the point of this badge.
- **Stations must be hands-on.** Two minutes of doing beats ten minutes of describing at every station.
- **Every Scout leaves with a step.** Don't let the pathway map end at "someday" — pin down one action with a when.

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Junior STEM Career Exploration badge; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.